ThermoBridge: Electrothermal Modelling and Fixed-Point Control of a Microhotplate

Amine Allab | Ishaq Kolla

Research technical report | 9 November 2026

ABSTRACT

Microhotplate control couples temperature-dependent electrical heating, spatial thermal nonuniformity, sampled resistance readout and finite-precision feedback. This work connects a two-dimensional thin-film finite-volume model, a positive-parameter thermal network and a synthesizable feedforward-plus-PI controller. Numerical validation includes analytical resistance, conservation, mesh refinement and independent pulse tests. A geometry study evaluates 27 geometries at three operating points and reserves nine whole geometries for surrogate testing. Actual RTL feedback experiments identify integral quantization and heating-rate effects. Joint gain and calibration selection at a 0.15 V heater supply leaves tracking errors above the objective in several cases. A prospective operating-point study holds the selected controller fixed and compares 0.15 V with 0.18 V on six fresh paired cases. At 0.18 V, hold-window RMSE spans 1.675-1.777 °C and worst hold overshoot is 1.297 °C. Peak heater power rises by 44%. An independent spatial pulse at the higher supply gives 0.128 °C compact-model RMS error and 0.324% time-step sensitivity. The routed SKY130 core passes the installed physical checks, with 10 MHz timing conditional on a 15 ns ADC arrival budget. The study uses assumed membrane properties and an ideal external sensing front end.

Index Terms- Microhotplates, electrothermal simulation, finite-volume methods, fixed-point arithmetic, feedback control, open-source ASIC design.

I. INTRODUCTION

Suspended microhotplates provide local temperature control with low thermal mass. Earlier fabricated systems combined heaters, sensing films and interface electronics [1]. Temperature-programmed operation also makes calibration stability and heterogeneous integration central engineering concerns [2]. Those demonstrations motivate the present control problem; their measured gas response, thermal performance and process capability are not results of this project.

ThermoBridge asks a narrower question: how do spatial-model reduction, resistance readout and finite arithmetic affect a controller that can be taken through an open digital implementation flow? The contribution is a reproducible integration study, rather than a new PI algorithm or a claim of superior sensor fabrication. Its evidence comprises a validated assumed-geometry model, recorded negative control results, actual RTL feedback experiments, and a routed digital core with explicit verification limits.

The external membrane, sensing front end, ADC and heater power switch are separate from the 1.8 V CMOS core. The baseline heater supply is 0.15 V and the proposed operating point uses 0.18 V. Control schedules span approximately 45-70 °C. The reference implementation uses IIC-OSIC-TOOLS [3], SKY130A [4] and LibreLane [5].

II. ELECTROTHERMAL AND READOUT MODELS

A. Spatial formulation

The suspended active region is 200 × 200 µm, with a 40 µm domain extension and four ambient-temperature anchors. The primary serpentine has an 8 µm trace width and 40 µm pitch; anchor width is 12 µm. Material properties, environmental losses and validity limits are explicit assumptions in materials.json and physics.json.

The electrical and thin-film thermal balances are

$$\nabla \cdot [\sigma(T)\nabla\phi] = 0 \tag{1}$$

$$C_A \frac{\partial T}{\partial t} = \nabla \cdot (K_A \nabla T) + q_J - q_{\text{loss}} \tag{2}$$

Here, the sheet heat capacity and conductance include the heater only where its mask is present. Harmonic face conductances enforce conservative interface fluxes. Anchors are fixed at ambient and suspended lateral boundaries are adiabatic. The sparse steady solution iterates electrical heating and thermal response. Transients use implicit thermal integration with an electrically lagged source, whose splitting error is assessed by time-step refinement.

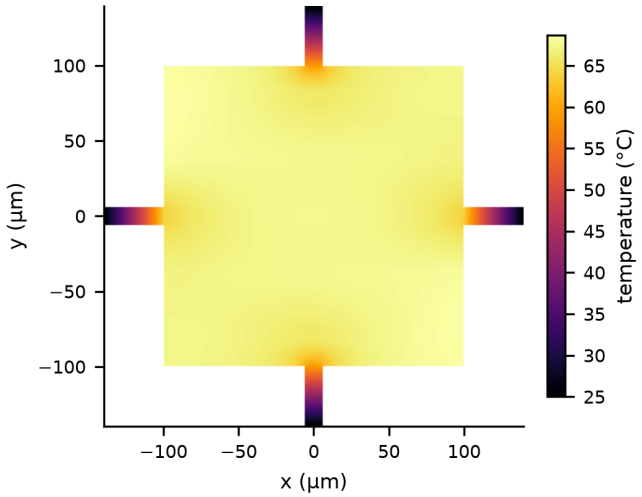


Fig. 1. Actual 1 µm-grid solution at 50% duty, with assumed geometry and boundary conditions. Field, masks and coordinates.

B. Compact plant and sampling

The reduced plant is

$$C_{\mathrm{th}}\dot{T} = P - \frac{T - T_a}{R_{\mathrm{th}}}, \qquad P = \frac{dV_h^2}{R_h(T)} \tag{3}$$

The positive fitted parameters are 349,137 K/W and 75.895 nJ/K, giving a 26.498 ms time constant. They describe the primary 4 µm spatial discretization, rather than a universal microhotplate. A separate 65%-duty pulse and cooldown test is reserved for compact-model evaluation.

Heater resistance follows a declared linear temperature coefficient. A nonlinear divider, external gain stage and 12-bit ADC model represent quantization, delay, gain/offset error and noise. Samples outside observable powered windows are invalid. A 1/256 heater probe at commanded-zero powered samples is included in heater energy; cooldown remains off. Divider-source energy is reported separately. Amplifier and digital switching power are unavailable.

III. CONTROLLER AND VERIFICATION METHOD

A. Implemented arithmetic

The RTL uses signed 16-bit temperature with six fractional bits, signed 18-bit coefficients with 14 fractional bits, a 24-bit integrator and a 26-bit combined sum. Rounding is nearest with ties to even; saturators and anti-windup bound state and output. A 16-entry, 9-bit feedforward table is configured through shadow registers and an atomic commit. The duty range 0-256 gives exact off/full endpoints for an eight-bit PWM phase counter.

The nominal 10 MHz clock produces 39.0625 kHz PWM and a 1 ms control interval. Stale data, rail values, invalid calibration, overtemperature and shutdown invoke latched safe-off behaviour. Fault clearing and reset are tested explicitly. Cocotb [6] compares real Icarus RTL outputs against the integer reference. Current checks comprise **29 Python tests and two directed/randomized RTL tests**. The v2 study contains 46 actual RTL feedback runs. The new calibration and supply studies add **71 runs**, including training and paired evaluation.

B. Integral quantization and frozen retuning

The implemented integral increment is quantized to whole Q8 duty counts:

$$\Delta I = \mathrm{round}_{\mathrm{even}}\left(\frac{K_{i,\mathrm{q14}} e_{\mathrm{q6}}}{4096}\right) \tag{4}$$

At a 1 °C error, the original coefficient 4 gives a zero increment. The new study therefore tests supported coefficient settings without changing RTL or invalidating its routed geometry. The frozen v2 protocol evaluates 12 coefficient pairs on three training schedules. Selection minimizes mean training hold RMSE; a selection file is written before evaluation starts.

Four evaluation cases introduce new setpoints, seeds and ambient/noise conditions. The previously observed 55→65 °C schedule is labelled a regression, not a new held-out test. Both baseline and retuned coefficients run on each identical case. The external ideal centred front end uses 18 mV excitation, an 8.6 mV reference, gain 800 and noiseless two-point calibration at 40/80 °C. Its energy is not a transistor-level amplifier measurement.

C. Geometry exploration

The spatial-screen protocol varies trace width over 8/12/16 µm, pitch over 32/40/48 µm and anchor width over 8/12/16 µm. Each actual mask is solved at duties 0.35, 0.50 and 0.65 on a 4 µm grid. A fixed split reserves nine entire geometries before solving; a regularized quadratic surrogate is fitted to the remaining 18. Held-out scores cannot influence candidate selection.

Training geometries that bracket a 55 °C mean are interpolated for power/nonuniformity screening. Selected points on that discrete training front are solved again on a 2 µm grid, with duty found by bisection and an independently executed 160 ms transient. This is a bounded steady-state screening exercise. It is not a closed-loop energy Pareto optimum, Bayesian-optimization benchmark or evidence of global optimality.

IV. RESULTS

A. Numerical validation

Table I. Numerical-model validation.

Quantity	Observed result	Interpretation
Rectangular resistance	0.655875 vs 0.6625 Ω	1% continuum/discrete difference
Peak at 4 / 2 / 1 µm	67.470 / 68.260 / 68.637 °C	0.55% final refinement change
Balance residuals	Below 1e-8 relative	Internal conservation, not measurement agreement
Compact held-out RMSE / maximum error	0.04427 / 0.06857 °C	Unseen pulse amplitude and duration
1 ms vs 0.5 ms transient	0.385% normalized difference	Recorded splitting/time-step sensitivity
Switched vs averaged RC PWM	0.00550 °C maximum endpoint difference	Three duties, 800 cycles each

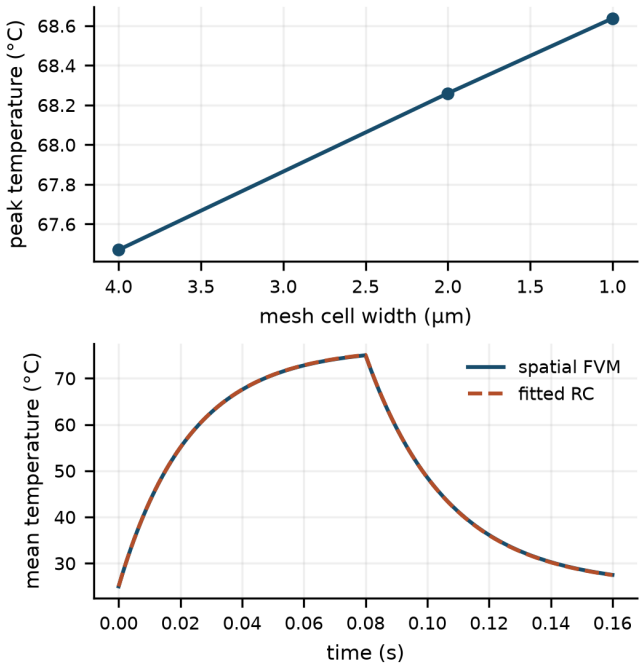


Fig. 2. Mesh refinement and held-out pulse response. Validation JSON, raw transient arrays, PWM check.

B. Paired reference campaign

The original Python campaign freezes 20 variation scenarios and three seeds across three floating controllers and six arithmetic profiles: 540 trials in total. Hold-window errors include the prescribed midpoint setpoint step; they are not errors measured only after settling. All settling failures remain in the aggregate.

Table II. Original paired reference campaign.

Controller	Mean hold RMSE (°C)	Mean heater energy (µJ)	Settling failures
Bang-bang	9.092	16.512	60/60
PI	9.544	14.786	57/60
Feedforward + PI	7.125	16.030	36/60

Feedforward improves mean tracking relative to this PI baseline, but uses more heater energy. These data do not establish the proposed energy saving. The Python width profiles share fractional precision and are not three routed circuit variants; their numerical results cannot establish an area advantage. Their accumulator model also differs from the exact primary RTL reference and is treated as a separate arithmetic study.

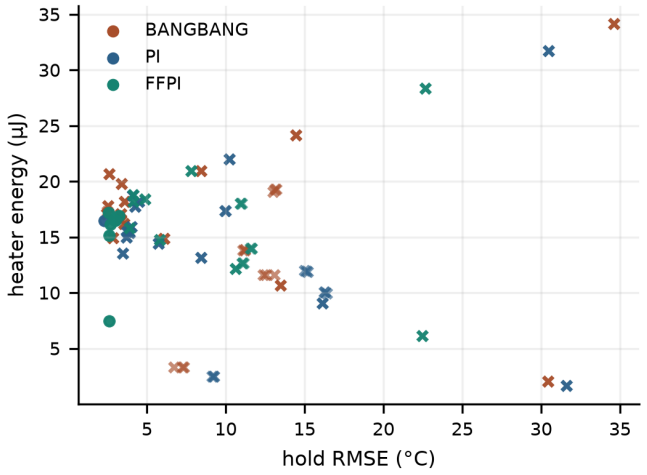


Fig. 3. Original paired campaign; crosses denote settling failures. Raw metrics, configuration, aggregate.

C. Actual RTL retuning

Training selects Kp=1147/16384 and Ki=32/16384 per sample, with mean training RMSE 2.054 °C. The baseline coefficients are recorded in the original study.

Table III. Paired actual-RTL evaluation.

Case	Baseline RMSE (°C)	Retuned RMSE (°C)	Evaluation role
new52	2.455	2.126	New evaluation
new57	2.537	2.214	New evaluation
new52 cold	2.977	2.476	New evaluation
new57 warm	1.967	2.006	New evaluation
legacy55 regression	2.342	2.101	Observed regression

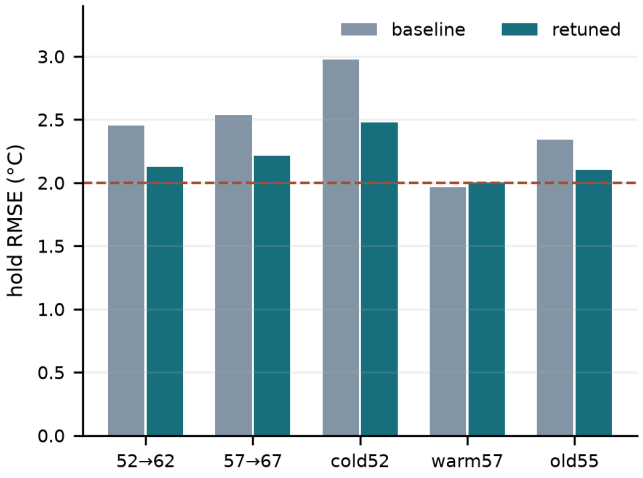


Fig. 4. Baseline and frozen retuned coefficients on identical cases. The dashed line is the 2 °C objective. Study summary; every row links to an actual RTL run through its recorded run ID.

Retuning improves three of the four new cases, but slightly worsens the warm-ambient case. Neither result set satisfies the 2 °C objective across the new cases. The regression improvement is useful diagnostic evidence, not a fresh independent success. Powered-phase faults are checked in every run; cooldown may latch stale while disabled PWM remains off. CONTROL_DIV is accelerated to 32 clocks while the plant advances by 1 ms per sample. This is actual RTL feedback, but not a simulation of every physical PWM period.

D. Calibration and heater operating point

The v3 joint study selects Kp=1638/16384, Ki=32/16384 per sample and calibration at 45/70 °C. Its mean training RMSE is 2.013 °C. At 0.15 V, five of six fresh v3 cases exceed 2 °C. The v4 protocol inherits these coefficients and compares heater supply on six further fresh cases and two regressions.

Table IV. Paired external heater operating points.

Fresh case	0.15 V RMSE (°C)	0.18 V RMSE (°C)	0.18 V hold overshoot (°C)
v4_E01	2.004	1.674	0.778
v4_E02	2.161	1.714	0.910
v4_E03	2.253	1.777	1.145
v4_E04_cold	2.166	1.677	0.596
v4_E05_warm	1.953	1.698	1.098
v4_E06_warm	1.943	1.744	1.297

All six 0.18 V cases meet the 2 °C hold-RMSE and 5 °C overshoot limits. Maximum measured hold overshoot is 1.297 °C. Powered-phase fault counts are zero. The higher supply raises available peak heater power by 44%, with the sensing excitation held at 18 mV. These results belong to the proposed 0.18 V operating point.

An independent 25 ms full-duty spatial pulse and cooldown at 0.18 V validates the original frozen compact coefficients. At dt=0.25 ms, RMS error is 0.128 °C and maximum absolute error is 0.226 °C. Halving the step changes the trajectory by 0.324%. The spatial peak reaches 99.173 °C within the declared material-model range. Both step sizes pass the predeclared numerical gates.

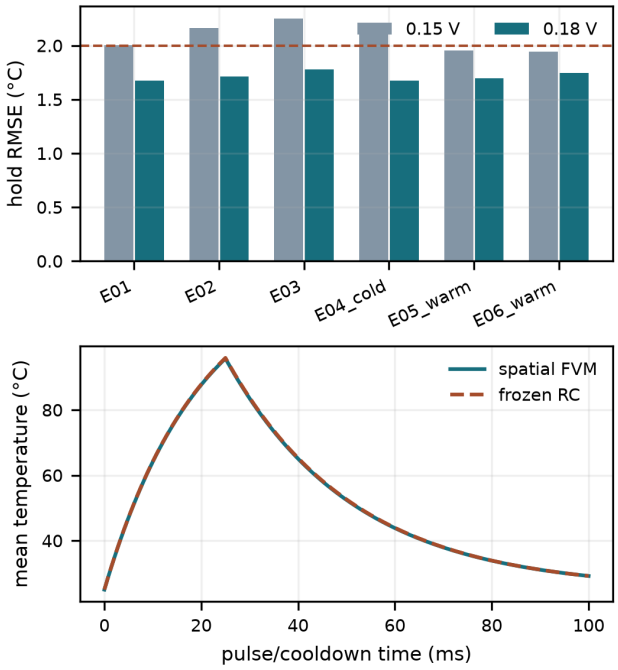


Fig. 5. Fresh paired RTL cases at two heater supplies and an independent full-duty spatial pulse at 0.18 V. Operating-point study, spatial validation, conditional heating reference.

The reference calculation starts at the pre-step setpoint, assumes perfect readout and applies full heating after the step. Its errors exceed 2 °C in four v3 cases at 0.15 V. Anticipatory trajectories and different temperatures at the step require separate analysis. The v4 experiment therefore declares a higher external supply before its fresh evaluation. Heater voltage is the paired treatment, with gains, calibration and control timing held equal between each pair.

E. Spatial geometry study

All 81 operating points solved with no recorded solver failures. Maximum electrical and thermal balance residuals are 5.20e-13 and 2.65e-08. On the nine held-out geometries, temperature prediction has median absolute error 2.124 °C and worst absolute error 5.133 °C. This error limits surrogate-only selection; final candidates require direct spatial solves. Two points lie on the discrete training front and are refined.

Table V. Direct 2 µm candidate validation at a 55°C target.

Candidate (width/pitch/anchor, µm)	Refined mean (°C)	Power (µW)	Central std. (°C)	Warmup to 54°C (ms)
G04 (8/40/12)	54.992	84.519	0.0964	86
G02 (8/32/16)	54.987	106.188	0.0874	70

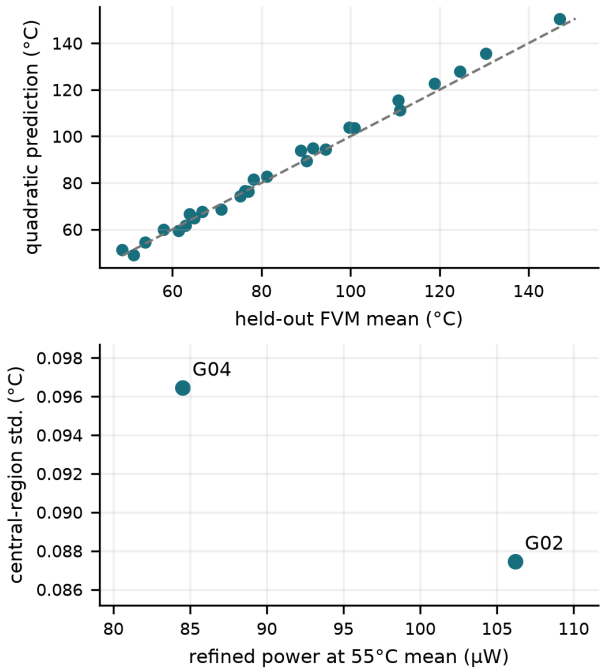


Fig. 6. Coarse training/test geometries and selected 2 µm candidate results. Summary, raw screen, split and fields. Finite-grid screening does not establish a global optimum.

F. Physical implementation

The implemented `sky130_fd_sc_hd` core reports 69,405.3 µm² standard-cell area and 158,379 µm² core area. Final GDS, DEF, LEF, mapped/powered netlists, constraints and SPEF are archived. Magic DRC, KLayout DRC and Netgen LVS report zero errors; detailed-routing and antenna checks are also zero. The v2 coefficient study changes registers, not this verified RTL or geometry.

Table VI. Conditional routed timing qualification.

Routed timing condition	Worst setup slack	Worst hold slack	Status
100 ns clock; 20 ns ADC arrival	-3.319 ns	+0.107 ns	Setup fails
Same geometry; 15 ns ADC arrival	+1.681 ns	+0.107 ns	Conditional pass

Other synchronous input/output budgets remain 20 ns. Nine installed corners are included in the recheck; no multicycle or disabled setup/hold paths are used. The conditional pass requires an external ADC source meeting the tighter arrival constraint. It is not measured maximum frequency. Slew/capacitance warnings remain, and installed open-PDK checks are not foundry-qualified signoff.

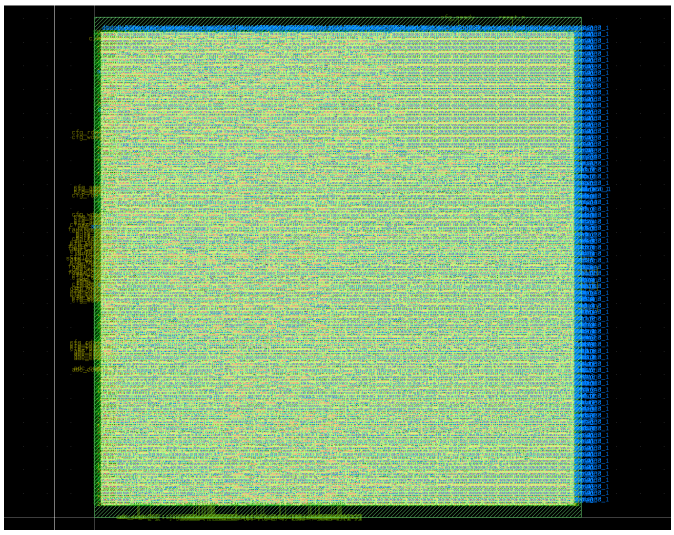


Fig. 7. KLayout [7] view of the actual generated GDS. Physical summary, original logs and checks, view extents.

V. DISCUSSION AND LIMITATIONS

Spatial approximation, readout calibration and controller arithmetic contribute different errors. Compact-model accuracy does not imply closed-loop accuracy, and RTL comparisons do not validate a physical sensor. The warm-ambient counterexample limits the generality of the integral retuning.

The membrane model is two-dimensional, uses assumed properties and lacks independent 3-D or measured validation. The readout amplifier remains ideal. Neither amplifier power nor activity-qualified digital-core power is available, so heater-terminal energy must not be presented as total-system energy. The new surrogate is tested on unseen geometries from a small assumed design grid, not on fabricated devices.

The current prototype includes spatial screening, actual RTL control and routed digital evidence. A full PFE extension still requires the separately laid-out transistor OTA, extracted analogue characterization, three physically implemented arithmetic variants, formal/gate-level checks and independent physical/model reproduction. No gas selectivity, certified temperature limit, fabricated silicon or complete mixed-signal chip is claimed. Detailed limits.

VI. CONCLUSION

ThermoBridge connects a conservative spatial thermal model, fixed-point control and a conditionally timed digital layout. The 0.18 V operating point meets the tracking and overshoot objectives on six fresh model-based cases. That result requires a higher external heater supply and 44% greater peak heater power. Historical 0.15 V failures remain in the record. Further engineering work includes a transistor sensing macro, activity-based power and independent physical verification.

VII. REPRODUCTION AND ARTIFACT AVAILABILITY

The environment record identifies the OSIC image digest, tool versions and PDK revision. Reuse that environment to reproduce the digital flow. The 30 October spatial study ran on the Windows host; its manifest records the Python, NumPy and SciPy versions separately. Historical run manifests retain their original source identifiers for reproducibility.

In PowerShell, from this checkout:

```
python scripts/container.py sync
python scripts/container.py run make test-physics test-
rtl
python scripts/container.py run python3
scripts/run_rtl_study.py
python scripts/container.py fetch
reports/rtl_study_v2.json results/rtl_study
python scripts/fetch_rtl_study.py
```

The new frozen studies and their selected raw runs can be reproduced with the commands below.

```
python scripts/container.py run python3
scripts/run_rtl_study.py --protocol
configs/rtl_study_v3.json
python scripts/container.py run python3
scripts/run_rtl_study.py --protocol
configs/rtl_study_v4.json
python scripts/container.py run python3
scripts/validate_operating_point.py
```

Fetch `reports/rtl_study_v3.json`, `reports/rtl_study_v4.json`, `results/rtl_study`, `reports/operating_point_validation.json` and `results/operating_point` after running. Use `scripts/fetch_rtl_study.py --report reports/rtl_study_v4.json` for the 19 v4 raw runs. `python scripts/release_check.py --operating-point` checks the declared 0.18 V objectives. The legacy `--strict` gate continues to expose the original 0.15 V failure.

`make simulate-operating-point` runs the frozen 19-run v4 study through its standard entry point. Fetch the new report and regenerate the article after a rerun so its evidence hashes identify the current run.

For the bounded spatial screen and article on a host with the documented dependencies:

```
python scripts/run_geometry_study.py
python scripts/build_article.py
python scripts/release_check.py
```

`make sweep` executes the bounded geometry study and `make report` builds this article. Physical implementation is explicit: `make synth` or `make gds`. `make verify-layout` audits archived hashes rather than rerunning DRC/LVS. The strict check still fails the tracking objective. Earlier clean-source reproduction, current verification and the original report preserve scope and results.

AUTHORS AND CONTRIBUTIONS

ThermoBridge is a collaborative research and engineering project developed by **Amine Allab** and **Ishaq Kolla**. The project combines electrothermal modelling, numerical verification, fixed-point control, RTL evaluation, geometry exploration and physical design. Both authors contributed to the project's development and documentation. Technical results, implementation details and reproducibility evidence are provided in the repository and its contribution record.

REFERENCES

[1] M. Y. Afridi *et al.*, “A monolithic CMOS microhotplate-based gas sensor system,” *IEEE Sensors Journal*, vol. 2, no. 6, pp. 644-655, Dec. 2002, doi: 10.1109/JSEN.2002.807780. Primary full text.

[2] J. Geist and M. Afridi, “Temperature-programmed gas-sensing with microhotplates: An opportunity to enhance microelectronic gas sensor metrology,” in *Frontiers of Characterization and Metrology for Nanoelectronics*, AIP Conf. Proc., vol. 1173, pp. 207-211, 2009. Primary full text.

[3] IIC, Johannes Kepler University, “IIC-OSIC-TOOLS,” software repository. Online.

[4] SkyWater Technology, “SkyWater SKY130 PDK documentation.” Online.

[5] LibreLane contributors, “LibreLane,” software repository. Online.

[6] Cocotb contributors, “Cocotb documentation.” Online.

[7] M. Köfferlein, “KLayout documentation.” Online.